

Subtopic: Arrays

Notes

DSA using Java

Contains 1 PDF lessons

Generated: 6 April 2026

KnowledgeGate

Module

Algorithm Concepts and Problems (Easy/Medium/Hard)

Topic

DSA by Prashant sir

Subtopic

Arrays

Lesson

ARRAY

Lesson 1 of 1

Array

And the award for
Best Answer Ever goes to

INTERMEDIATE-II

Computer programming

1. Define Array.

Ans: * array:

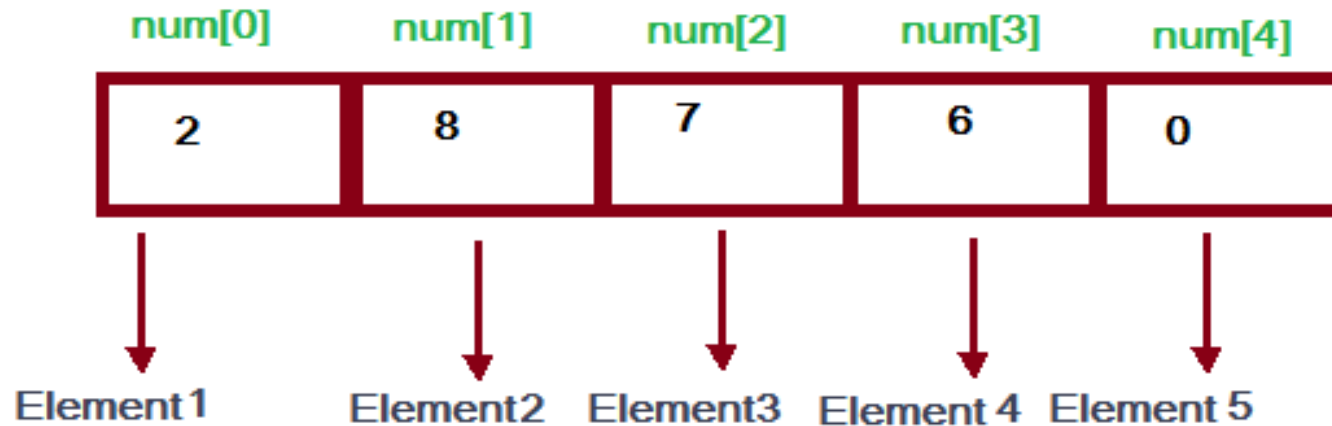
An Array is used to call a boy or a person who is at a distance far away from us who are visible to our naked eye.

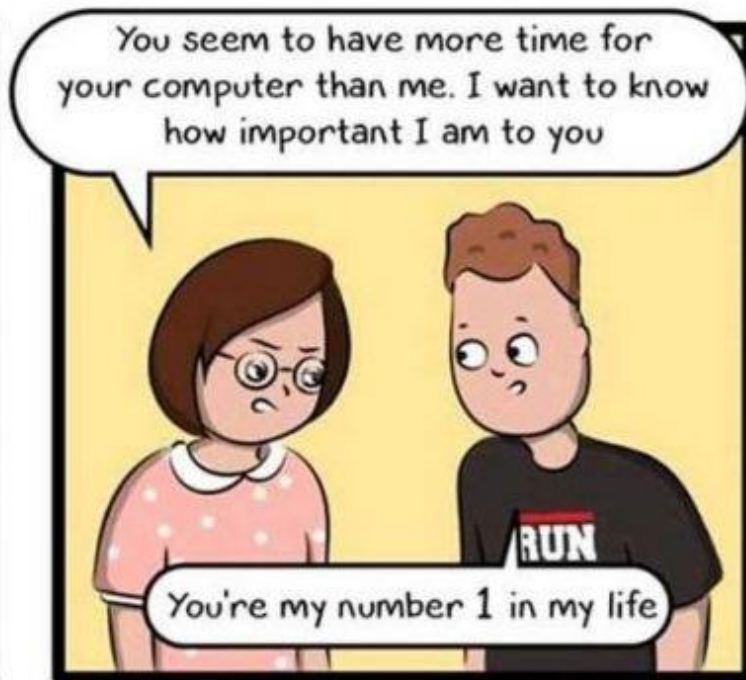
Example: Array Rupesh.

met me

Array

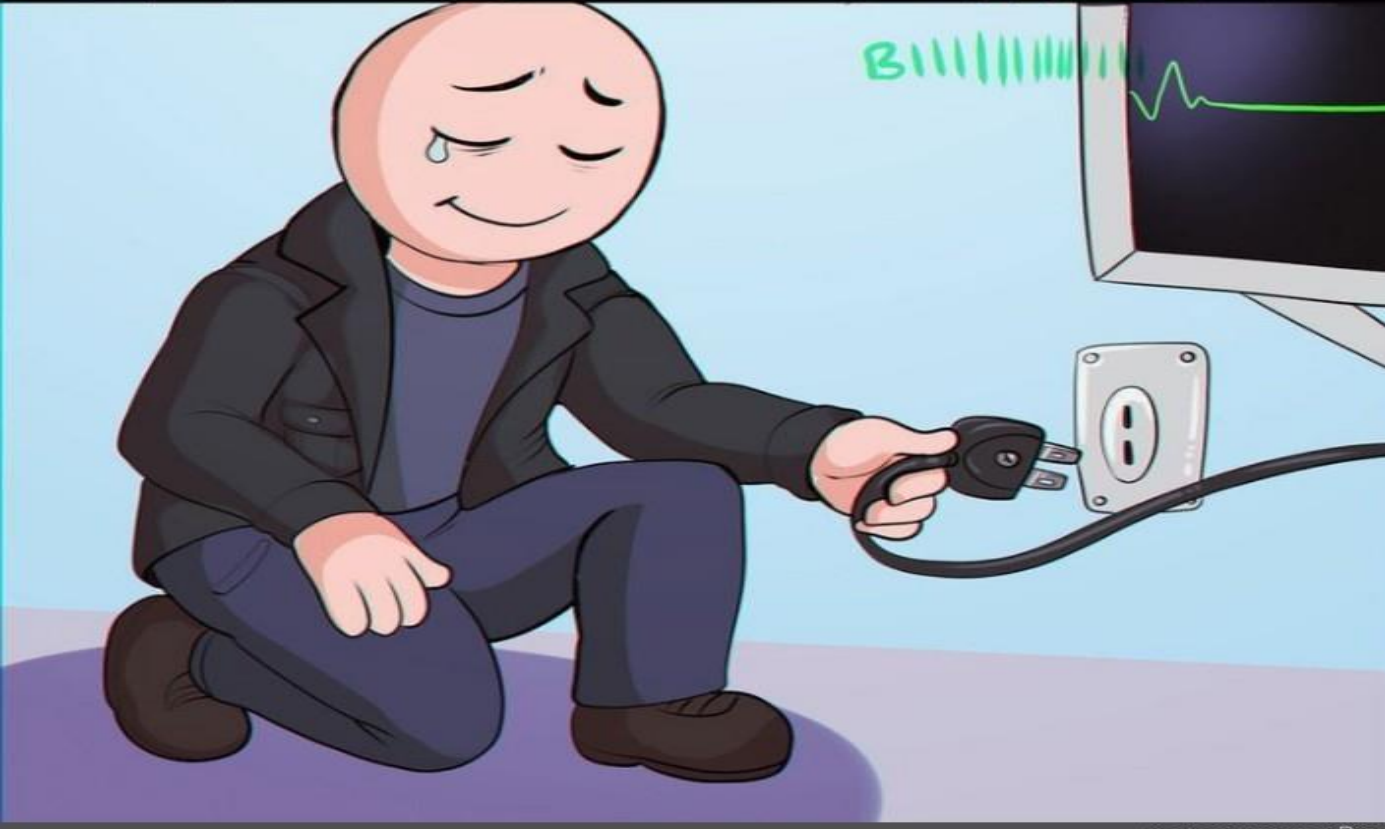
- An array is a collection of items stored at contiguous memory locations.
- The idea is to store multiple items of the same type together. This makes it easier to calculate the position of each element by simply adding an offset to a base value, i.e., the memory location of the first element of the array (generally denoted by the name of the array).
- Each element can be uniquely identified by their index in the array.





Size of an array

- Number of elements = (Upper bound – Lower Bound) + 1
 - Lower bound index of the first element of the array
 - Upper bound index of the last element of the array



- | Memory Location | | | | | | | | | |
|-----------------|-----|-----|-----|-----|-----|-----|---|---|---|
| 200 | 201 | 202 | 203 | 204 | 205 | 206 | • | • | • |
| U | B | F | D | A | E | C | • | • | • |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | • | • | • |
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Break



**HOW DO I CREATE A MULTIDIMENSIONAL
ARRAY FROM A SINGLE ARRAY?**



Two-Dimensional array

- The two-dimensional array can be defined as an array of arrays. The 2D array is organized as matrices which can be represented as the collection of rows and columns.
- However, 2D arrays are created to implement a relational database look a like data structure. It provides ease of holding the bulk of data at once which can be passed to any number of functions wherever required.

	Column 0	Column 1	Column 2
Row 0	x[0][0]	x[0][1]	x[0][2]
Row 1	x[1][0]	x[1][1]	x[1][2]
Row 2	x[2][0]	x[2][1]	x[2][2]

Two-Dimensional array

